

Quantitative Market Signal Research & Portfolio Risk Analysis

Out-of-sample BTCUSDT directional prediction, trading-cost analysis, robustness testing and risk evaluation

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Question. Can information available at time t predict the direction of the next 60-minute BTCUSDT return out of sample, well enough to beat no-skill *and* make money after realistic trading costs? **Answer:** the direction is slightly but robustly predictable; the edge is far too small to pay for trading it.

0.545 test AUC (95% CI 0.538–0.552)	p = 0.004 log-loss gain vs base rate (Newey–West)	−5.84 net Sharpe (CI −6.94 to −4.77)	0.19 bp break-even cost per side vs 7 bp assumed	23 / 23 robustness variants negative after costs
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Data

Binance Vision spot BTCUSDT 1-minute candles, 17 Aug 2017 – 30 Sep 2026: **4,789,279 rows** from 110 monthly files, every one SHA-256 verified; ETHUSDT and perpetual funding rates as secondary data. The audit found no duplicates or impossible candles, 0.63% missing minutes (none in 2024–26), Binance's 2025 switch from millisecond to microsecond timestamps, and 21,602 candles shifted 20.8 s / 14.8 s off the minute. Those were treated as missing, because snapping them to the grid would leak future seconds into "closed" features.

Method

- Design, parameters and verdict rule **committed to git before any data was downloaded**.
- Decide on the hour from closed candles only; enter at the next minute's open; exit 60 minutes later. Label = sign of that log return; labels never overlap.
- 16 scale-free features (returns, momentum, realised volatility, relative volume, taker-buy imbalance, candle position). Randomising all data after t at 24 real times changed **0 of 16** features.
- Logistic regression vs small LightGBM; purged expanding walk-forward over 10 half-years (2019–23); LightGBM kept only if it raised mean AUC by > 0.005 *and* beat LR on log-loss in $\geq 70\%$ of folds (it passed at exactly 7/10).
- Selection frozen, then **one untouched test**, Jan 2024 – Sep 2026, with six-monthly refits.

What the model learned

A short-term **reversal**: after a rise, the next hour is slightly more likely to be weaker. It replicated on ETHUSDT with no re-tuning (AUC 0.546). Accuracy was 53.5% against a 50.5% base rate, but mean next-hour returns across probability deciles span only -1.3 to $+1.9$ bp.

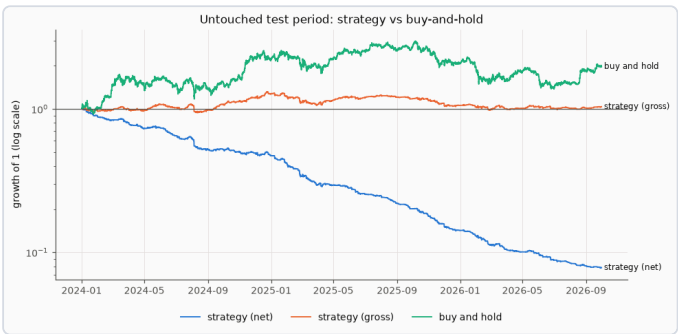
Conclusion. The study found a small, persistent and statistically detectable reversal signal in BTCUSDT, and showed that it is not economically viable for a taker at realistic costs. Predictive accuracy did not become profit. Pre-registered verdict: **weak / inconclusive**; the strategy should not be deployed.

Relevance to quantitative research at Gravia

Hypothesis testing with pre-registered pass/fail criteria · ML that turns probabilities into positions and costed trades · scepticism toward backtests (leakage test, purged walk-forward, one test run, every trial logged) · messy multi-million-row crypto data · understanding why a signal fails and decays. *Not covered:* prediction-market order books or fills, on-chain/wallet data, SQL/Polars, live trading.

Trading result (untouched test)

	Strategy	Buy & hold
Cost model	7 bp/side (5 bp fee + 1 bp spread + 1 bp slippage, last two assumed) + actual funding	
Net return	−92.2%	+97.6%
Net / gross Sharpe	−5.84 / 0.16	0.76
Max drawdown	−92.3%	−53.7%
Time in market; round trips/yr	9.1%; 667	100%



Test period, log scale: strategy net and gross of costs vs buy-and-hold (figures/test_equity.png).

Robustness and risk

23 pre-specified variants (thresholds, feature removal, model swap, 0/5-min execution lag, shifted hours, 15/30-min horizons): AUC 0.534–0.548 in every one, net Sharpe negative in every one. Only zero costs are profitable; a spot-fee variant gives -9.86 . Daily 99% VaR 3.53% vs 2.18% under normality; rolling VaR passes Kupiec ($p = 0.42, 0.60$); 5,000 block-bootstrap years: none profitable. AUC fell from 0.567 (2024H2) to 0.532 (2026H2) as volatility features drifted (PSI up to 3.5).